Accuracy Report

AI Student Evaluator

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Version: 1.0
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AI Model: Google Gemini 2.5 Flash
Application: AI Student Evaluator
Date: 04 July 2026

1. Purpose

The purpose of this Accuracy Report is to demonstrate the reliability, consistency, and effectiveness of the **AI Student Evaluator** in assessing students' understanding of classroom sessions.

Unlike traditional assessment systems that rely on predefined keywords or manually created rubrics, this solution leverages **Google Gemini 2.5 Flash** to perform semantic analysis by comparing the mentor's reference summary against students' WhatsApp discussions.

The report validates that the generated evaluations—including scores, strengths, missing concepts, mentor recommendations, and learning goals—accurately reflect each student's level of understanding while maintaining consistency across the complete evaluation workflow.

1.1 Objectives

The primary objectives of this report are:

- Evaluate the accuracy of AI-generated student assessments.
- Verify that AI-generated scores align with students demonstrated understanding.
- Validate the identification of strengths and knowledge gaps.
- Assess the relevance and usefulness of mentor recommendations.
- Demonstrate the reliability of the complete evaluation pipeline from mentor summary ingestion to dashboard generation.
- Provide evidence that the solution satisfies the competition objective of analysing **"What Was Taught"** versus **"What Was Understood."**

1.2 Scope

This report covers the complete AI evaluation process, including:

- Mentor Summary Preparation (Tactiq.ai Transcript)
- WhatsApp Chat Processing
- Student Message Parsing

- Google Gemini AI Evaluation
- Class Analytics Generation
- React Dashboard Validation
- PDF Report Generation

The report focuses on the accuracy of AI-generated evaluations rather than infrastructure or performance benchmarking.

2. Evaluation Methodology

2.1 Overview

The AI Student Evaluator follows a structured evaluation methodology that combines transcript analysis, student discussion analysis, Large Language Model (LLM)-based semantic comparison, and analytics generation to determine each student's understanding of a classroom session.

Instead of relying on traditional keyword matching, the solution compares the concepts explained by the mentor with the concepts recalled and discussed by students. This enables the system to evaluate conceptual understanding, identify learning gaps, and generate personalized feedback for each student.

2.2 Evaluation Workflow

Classroom Session

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Tactiq.ai Meeting Transcript

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Mentor Reference Summary

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WhatsApp Group Discussion   Student Chat Parser

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Student-wise Conversations
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        ▼
Prompt Construction Engine
        │
        ▼
Google Gemini 2.5 Flash AI
        │
        ▼
Structured Student Evaluation
        │
        ▼
┌───────────────────────────────┐
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▼                ▼
Class Analytics      Individual Evaluation
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        ▼
React Analytics Dashboard
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JSON Export & PDF Report Generation
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2.3 Evaluation Process

The evaluation is performed in the following sequence:

1. **Mentor Summary Preparation**
   The classroom transcript generated using **Tactiq.ai** is converted into a structured mentor summary that serves as the reference knowledge base for the session.

2. **Student Discussion Collection**
   Students' WhatsApp discussions are provided as plain text input and grouped by participant.

3. **Student-wise Parsing**
   Multiple messages from the same student are merged to create a complete representation of that student's understanding.

4. **AI-based Semantic Evaluation**
   The mentor summary and each student's discussion are submitted to **Google Gemini 2.5 Flash**, which performs semantic comparison and generates:

   - Understanding Score
   - Strengths
   - Missing Concepts
   - Mentor Recommendation
   - Next Learning Goal
   - Overall Feedback

5. **Analytics Generation**
   Individual evaluations are aggregated to calculate class-level insights such as average score, highest score, lowest score, student rankings, and students requiring additional attention.

6. **Visualization & Reporting**
   The processed data is displayed through the React dashboard and can be exported as JSON or PDF reports for mentors.

3. AI Accuracy Evaluation Criteria

Parameter	**Description**	**Validation Criteria**
Concept Coverage	Measures how well the student covered the concepts discussed by the mentor.	AI correctly identifies covered and missed concepts.
Score Appropriateness	Evaluates whether the assigned score reflects the student's understanding.	Higher understanding should result in a higher score, while incomplete or incorrect understanding should receive proportionally lower scores.
Strength Identification	Validates the AI's ability to recognize concepts the student understood correctly.	Generated strengths should match the student's demonstrated knowledge.
Missing Concept Detection	Ensures the AI identifies concepts that were taught but not reflected in the student's discussion.	Missing concepts should align with the mentor summary.

Mentor Recommendation	Evaluates whether the generated recommendation is relevant and actionable.	Recommendation should address the student's identified learning gaps.
Next Learning Goal	Verifies that the AI suggests an appropriate next topic or concept for improvement.	Learning goal should naturally follow the identified knowledge gaps.
Feedback Quality	Measures the usefulness and clarity of the overall feedback.	Feedback should be consistent with the assigned score and evaluation.

3.1 Evaluation Principles

The following principles were adopted while validating AI-generated responses:

- The evaluation focuses on **semantic understanding** rather than exact keyword matching.
- Minor variations in wording are considered acceptable if the meaning remains unchanged.
- AI responses are expected to remain logically consistent across repeated evaluations.
- Recommendations should be concise, actionable, and directly related to the student's knowledge gaps.
- Feedback should accurately reflect both the student's strengths and areas for improvement.

4. AI Validation Methodology

4.1 Validation Approach

The AI evaluation was validated using the following approach:

1. **Reference Baseline Creation**
    - The mentor summary generated from the Tactiq.ai transcript was used as the reference knowledge base.
2. **Student Response Analysis**
    - Student discussions collected from WhatsApp were grouped and consolidated into individual responses.
3. **AI Evaluation**
    - Each student's discussion was evaluated independently using Google Gemini 2.5 Flash.

4. **Output Verification**
    - The generated score, strengths, missing concepts, recommendations, learning goals, and feedback were manually reviewed against the mentor summary.
5. **Consistency Review**
    - Multiple student responses representing different understanding levels were evaluated to verify consistent behaviour across the application.

4.2 Validation Scenarios

Scenario	Expected Behaviour
Student covers all major concepts	High score with minimal missing concepts
Student covers only key concepts	Moderate score with identified knowledge gaps
Student covers very few concepts	Low score with improvement recommendations
Student includes incorrect understanding	AI identifies incorrect or missing concepts
Student provides minimal response	Low evaluation score with additional learning guidance

5. Test Dataset & Evaluation Scenarios

5.1 Dataset Summary

Dataset	Source	Purpose
Mentor Summary	Tactiq.ai Transcript	Reference knowledge base for AI evaluation
Student Discussions	WhatsApp Group Chat	Measure students' understanding of the session
AI Evaluation Output	Google Gemini 2.5 Flash	Generate scores and personalized feedback
Dashboard Data	Backend Evaluation Engine	Generate class analytics and reports

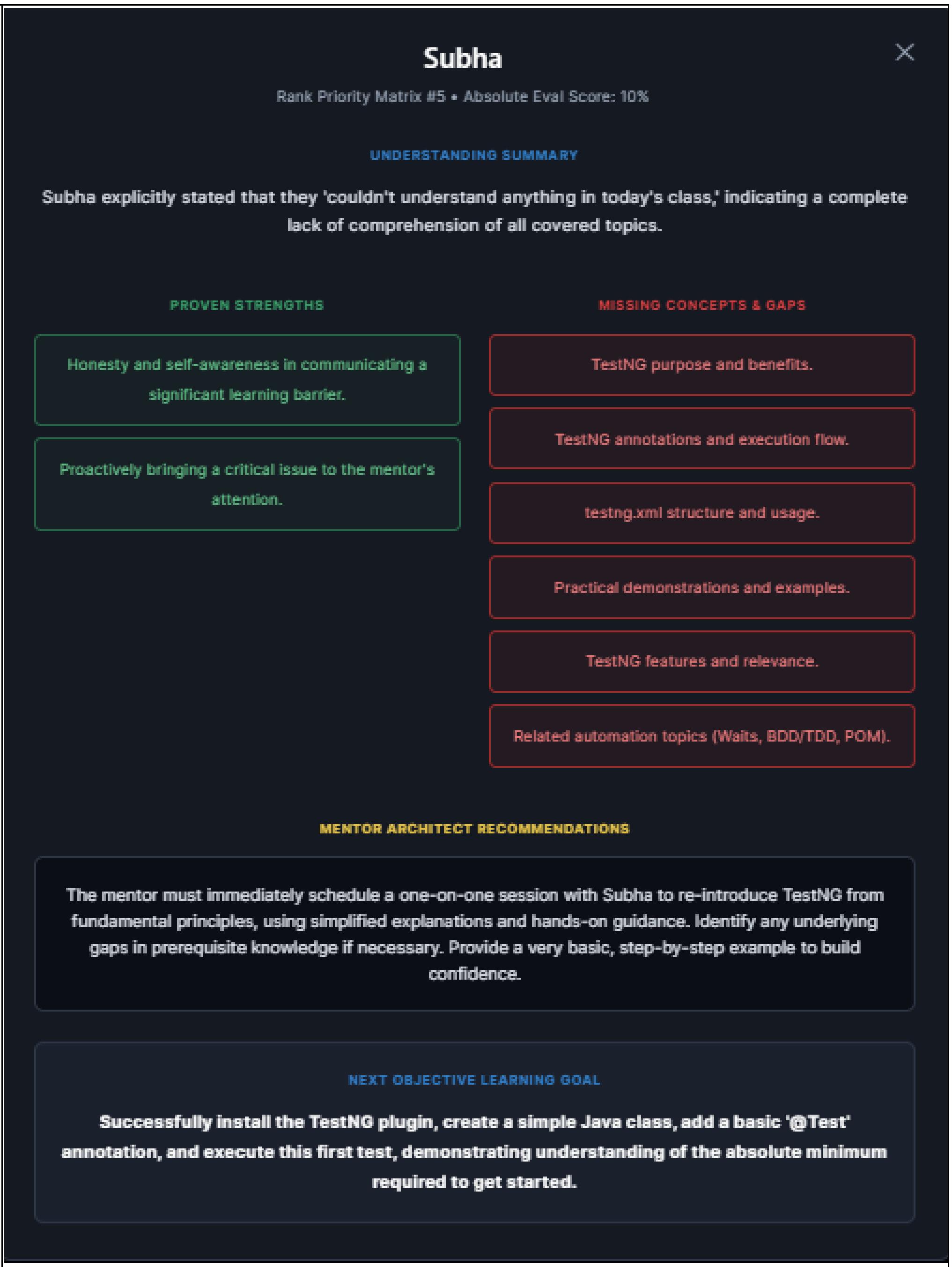
5.2 Evaluation Scenarios

Scenario	**Description**	**Expected Outcome**
Complete Understanding	Student captures most concepts discussed by the mentor	High score with minimal missing concepts
Partial Understanding	Student explains only a subset of the concepts	Moderate score with identified knowledge gaps
Limited Understanding	Student recalls very few concepts	Low score with improvement recommendations
Incorrect Understanding	Student includes incorrect or unrelated concepts	AI identifies inaccuracies and assigns an appropriate score
Multiple Student Evaluation	Several students participate in the same discussion	Independent evaluation generated for each student

6. AI Evaluation Results & Observations

6.1 AI Evaluation Results

Student	**AI Score**	**Observation**

<table>
<tr>
<td>Subha</td>
<td>10%</td>
<td>


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Aditi	85%	**Aditi** Rank Priority Matrix #1 • Absolute Eval Score: 85% UNDERSTANDING SUMMARY Aditi demonstrated a strong grasp of logistical details, recalling the requirement to submit a GitHub repository with a proper README and accurately listing the topics for the next class (advanced annotations, utilities, WebDriver Manager, and Page Object Model). PROVEN STRENGTHS Accurate recall of submission format and requirements. Clear understanding of upcoming learning topics. Attention to detail regarding class logistics and future planning. MISSING CONCEPTS & GAPS Did not discuss core TestNG concepts, annotations, or execution flow. Did not mention the practical aspects of the immediate TestNG assignment or the prior AI agent assignment. Limited to logistical and future-oriented information. MENTOR ARCHITECT RECOMMENDATIONS Encourage Aditi to actively prepare for the next session by researching the upcoming topics (e.g., Page Object Model) and formulating questions based on how they might integrate with TestNG or current automation practices. NEXT OBJECTIVE LEARNING GOAL **Research the Page Object Model (POM) and its advantages, identifying how it complements TestNG for UI automation, and come prepared with specific questions for the next session.**

Hunaid	**75%**	

6.2 Overall Observation

The evaluation results demonstrate that the AI consistently distinguished between different levels of student understanding. Students who covered a broader range of concepts received higher scores, while those with partial or limited understanding received proportionally lower scores along with targeted recommendations.

This behaviour indicates that the AI evaluates **conceptual understanding** rather than simply matching keywords, making the assessment more meaningful for mentors.

7. Accuracy Analysis & Key Findings

7.1 Key Findings

Observation Area	**Finding**
Concept Identification	The AI consistently identified concepts covered by students and highlighted missing topics based on the mentor summary.
Score Assignment	Students demonstrating stronger conceptual understanding consistently received higher evaluation scores, while incomplete discussions resulted in proportionally lower scores.
Strength Identification	The AI accurately highlighted concepts that students understood well, enabling mentors to recognize learning achievements.
Knowledge Gap Detection	Missing concepts were clearly identified, helping mentors focus on areas requiring reinforcement.
Mentor Recommendations	Recommendations were concise, relevant, and aligned with the student's identified learning gaps.
Feedback Quality	Generated feedback reflected the student's overall performance and provided meaningful guidance for improvement.

7.2 Strengths of the Evaluation Approach

- Performs **semantic evaluation** instead of simple keyword matching.
- Generates individualized feedback for each student.
- Identifies both strengths and areas requiring improvement.
- Produces structured outputs suitable for dashboards and reports.
- Reduces the manual effort required to review lengthy classroom discussions.

7.3 Observed Limitations

While the evaluation produced reliable results, the following factors may influence AI-generated outputs:

- The quality of the evaluation depends on the completeness of the mentor reference summary.
- Student responses with very limited information may result in less detailed feedback.
- As with any Large Language Model, minor variations in wording may occur between executions.

- The AI is intended to assist mentors and should complement, rather than replace, human academic judgment.

8. Conclusion & Future Enhancements

8.1 Conclusion

The **AI Student Evaluator** successfully demonstrates how Artificial Intelligence can assist mentors in assessing students' understanding of classroom sessions in an efficient, structured, and scalable manner.

By comparing the **mentor's reference summary** with **students' WhatsApp discussions**, the application generates semantic evaluations that include performance scores, strengths, missing concepts, mentor recommendations, learning goals, and class-level analytics.

The validation performed throughout this report indicates that the AI consistently differentiates varying levels of student understanding while providing meaningful and actionable insights. The structured evaluation process, combined with an interactive React dashboard and PDF reporting, significantly reduces the manual effort required for classroom assessment and enables mentors to make informed decisions more efficiently.

The solution is intended to serve as an intelligent decision-support tool, complementing the mentor's expertise rather than replacing human judgment.

8.2 Future Enhancements

Area	**Proposed Enhancement**
Input Sources	Support PDF, DOCX, PowerPoint, and LMS content in addition to text input.
Student Platforms	Integrate directly with Microsoft Teams, Google Classroom, Slack, or WhatsApp Business APIs.
AI Evaluation	Compare results across multiple LLMs (Gemini, GPT, Claude) for improved consistency and benchmarking.
Analytics	Introduce trend analysis, learning progress tracking, and historical performance dashboards.
Notifications	Automatically notify mentors about students requiring additional attention.
Authentication	Add secure login and role-based access for mentors and administrators.
Cloud Deployment	Deploy the application using cloud services for multi-user access and centralized data management.
Multilingual Support	Extend evaluation capabilities to support multiple languages and regional classrooms.

9. Final Statement

The AI Student Evaluator demonstrates the practical application of Generative AI in the education domain by transforming classroom discussions into meaningful learning insights. Through semantic evaluation, automated analytics, and intuitive reporting, the solution empowers mentors to monitor student understanding more effectively, provide personalized guidance, and enhance the overall learning experience.

The project showcases how AI can be responsibly integrated into educational workflows to improve productivity while preserving the mentor's role as the final decision-maker.

9.1 Report Summary

Report Attribute	**Details**
Report Name	Accuracy Report
Project	AI Student Evaluator
AI Model	Google Gemini 2.5 Flash
Input Sources	Tactiq.ai Transcript & WhatsApp Chat
Output	Student Evaluations, Dashboard Analytics, PDF Report
Report Version	1.0
Status	Final